

CTMC – Time spent in a state

- ▶ For a time homogeneous CTMC, we have a transition probability matrix with entries $p_{ij}(t)$, i.e., $P(t) = [[p_{ij}(t)]]$.
- ▶ Let $Y_t = \inf\{s > 0 : X(t+s) \neq X(t)\}$
- ▶ Y_t is the remaining time that the process spends in whichever state it is in, at time t .
- ▶ Intuitively, the time spent in a state should depend only on what state it is in, and not on the previous state.

Theorem

$$P(Y_t > u | X(t) = i) := \bar{G}_i(u) = e^{-a_i u}$$

for all $i \in \mathcal{S}$ and $t \geq 0, u \geq 0$ and for some real number $a_i \in [0, \infty]$.

Proof 1

- ▶ $\bar{G}_i(u + v) = P(X(s) = i, s \in [t, t + u + v] | X(t) = i)$
- ▶ $\bar{G}_i(u + v) = P(X(s) = i, s \in [t + u, t + u + v]; X(p) = i, p \in [t, t + u] | X(t) = i)$
- ▶ $P(AB|C) = P(A|BC)P(B|C)$
- ▶ Due to Markov property we have $P(AB|C) = P(A|B)P(B|C)$
- ▶ $P(X(s) = i, s \in [t + u, t + u + v] | X(p) = i, p \in [t, t + u]) =$
- ▶ $P(X(s) = i, s \in [t + u, t + u + v] | X(t + u) = i) = \bar{G}_i(v)$
- ▶ $P(X(p) = i, p \in [t, t + u] | X(t) = i) = \bar{G}_i(u)$
- ▶ $\bar{G}_i(u + v) = \bar{G}_i(u)\bar{G}_i(v)$
- ▶ Only CCDF function which satisfies this equation is the exponential distribution. This requires a proof. We will skip this part.

Simpler Proof

- ▶ Let τ_i denote the time the CTMC spends in state i before moving out. Suppose the CTMC is in state i at time 0.
- ▶ What is $P(\tau_i > s + t | \tau_i > s)$?
- ▶ Note that $X(s) = i$ and therefore from the Markov property,

$$\begin{aligned}P(\tau_i > s + t | \tau_i > s) &= P(X(u) = i, u \in [s, s + t] | X(t) = i, t \in [0, s]) \\&= P(X(u) = i, u \in [s, s + t] | X(s) = i) \\&= P((X(u) = i, u \in [0, t] | X(0) = i) \\&= P(\tau_i > t).\end{aligned}$$

- ▶ Since $P(\tau_i > s + t | \tau_i > s) = P(\tau_i > t)$, this implies the distribution has memoriless property and must be exponential.

Finite dimensional distributions

- ▶ Consider a DTMC $\{X_n, n \geq 0\}$ with tpm denoted by P .
- ▶ We assume M states and X_0 denotes the initial state.
- ▶ You can start in any starting state or may pick your starting state randomly.
- ▶ Let $\bar{\mu} = (\mu_1, \dots, \mu_M)$ denote the initial distribution.
- ▶ How does one obtain the finite dimensional distribution $P(X_0 = x_0, X_1 = x_1, \dots, X_k = x_k)$?

Finite dimensional distributions

- ▶ Consider a CTMC $\{X_t, t \geq 0\}$ with t-time pm given by $P(t)$.
- ▶ We assume M states and X_0 denotes the initial state.
- ▶ Let $\bar{\mu} = (\mu_1, \dots, \mu_M)$ denote the initial distribution.
- ▶ How does one obtain the finite dimensional distribution $P(X_0 = x_0, X_{t_1} = x_1, \dots, X_{t_k} = x_k)$?

Chapman Kolmogorov Equations for DTMC

- ▶ $P = [[p_{ij}]]$ denotes the one step transition probability matrix.
- ▶ Let $P^{(n)}$ denote the n-step transition probability matrix.
- ▶ CK equation tells us that $P^{(n+l)} = P^{(n)}P^{(l)}$.
- ▶ $p_{ij}^{(n+l)} = P(X_{n+l} = j | X_0 = i) = \sum_k P(X_{n+l} = j, X_n = k | X_0 = i)$
- ▶ $p_{ij}^{(n+l)} = \sum_k P(X_{n+l} = j | X_n = k, X_0 = i) P(X_n = k | X_0 = i)$
- ▶ $p_{ij}^{(n+l)} = \sum_k P(X_{n+l} = j | X_n = k) P(X_n = k | X_0 = i)$
- ▶ $p_{ij}^{(n+l)} = \sum_k p_{ik}^{(n)} p_{kj}^{(l)} = [P^{(n)}P^{(l)}]_{ij}$
- ▶ At which step did we use time homogeneity and the Markov property?

n step transition probabilities

- ▶ $P = [[p_{ij}]]$ denotes the one step transition probability matrix.
- ▶ Let $P^{(n)}$ denote the n-step transition probability matrix.
- ▶ From the CK equation we know that $P^{(n+l)} = P^{(n)}P^{(l)}$.
- ▶ It is easy to see that $P^{(n)} = P^{(n-1)}P$.
- ▶ For an M state DTMC, $p_{ij}^{(2)} = \sum_{k=1}^M p_{ik}p_{kj}$.
- ▶ This implies that the n-step transition probability matrix can be obtained as $P^{(n)} = P^n$
- ▶ Given X_0 and P , you can generate n-step probabilities or $P_{X_0}(X_n)$

Chapman Kolmogorov Equations for CTMC

- ▶ Let $P(t)$ denote the t-time transition probability matrix.
- ▶ CK equation for a CTMC is $P(t + l) = P(t)P(l)$.
- ▶ $p_{ij}(t + l) = P(X(t + l) = j | X(0) = i)$
- ▶ $= \sum_k P(X(t + l) = j, X(t) = k | X(0) = i)$
- ▶ $= \sum_k P(X(t + l) = j | X(t) = k, X(0) = i) P(X(t) = k | X(0) = i)$
- ▶ $= \sum_k P(X(t + l) = j | X(t) = k) P(X(t) = k | X(0) = i)$
- ▶ $p_{ij}(t + l) = \sum_k p_{ik}(t) p_{kj}(l) = [P(t)P(l)]_{ij}$